PYTHON ACTIVITY BASED LEARNING

PROBLEM--9

Repeated Transaction Detection: Process transaction identifiers and amounts in a list, detect duplicate transactions, identify threshold violations, and generate a list of suspicious entries

Objective:

To develop a Repeated Transaction Detection System that processes transaction identifiers and amounts, identifies duplicate transactions, detects transactions exceeding a predefined amount threshold, and generates a list of suspicious entries to support accurate transaction monitoring and prevent potential financial irregularities.

Algorithm:

1. Start.

2.Input the number of transactions.

3.For each transaction, input the transaction ID and amount.

4.Input the transaction amount threshold.

5.Initialize three lists: transactions, seen, and suspicious.

6.Process each transaction:

- Initialize an empty list of reasons.
- If the transaction ID already exists in seen, mark it as a duplicate.
- Otherwise, add the transaction ID to seen.
- If the transaction amount exceeds the threshold, mark it as a threshold violation.

7.If any suspicious reason exists, add the transaction details and reasons to suspicious.

8.Display all suspicious transactions with their IDs, amounts, and reasons.

9.Display the total number of transactions and suspicious entries.

10.Stop.

Pseudocode:

BEGIN

INPUT number_of_transactions

```
INITIALIZE transactions = empty list
FOR i = 1 TO number_of_transactions
    INPUT transaction_id
    INPUT transaction_amount
    ADD (transaction_id, transaction_amount)
        TO transactions
END FOR
INPUT threshold
INITIALIZE seen = empty list
INITIALIZE suspicious = empty list
FOR each transaction in transactions
    GET transaction_id and amount
    INITIALIZE reasons = empty list
    IF transaction_id IN seen THEN
        ADD "Duplicate transaction" TO reasons
    ELSE
        ADD transaction_id TO seen
    END IF
    IF amount > threshold THEN
        ADD "Threshold violation" TO reasons
    END IF
    IF reasons is NOT empty THEN
        ADD transaction details and reasons
            TO suspicious
    END IF
END FOR
DISPLAY "Suspicious Transactions"
FOR each item in suspicious
    DISPLAY transaction ID
```

    DISPLAY transaction amount

    DISPLAY suspicious reasons

END FOR

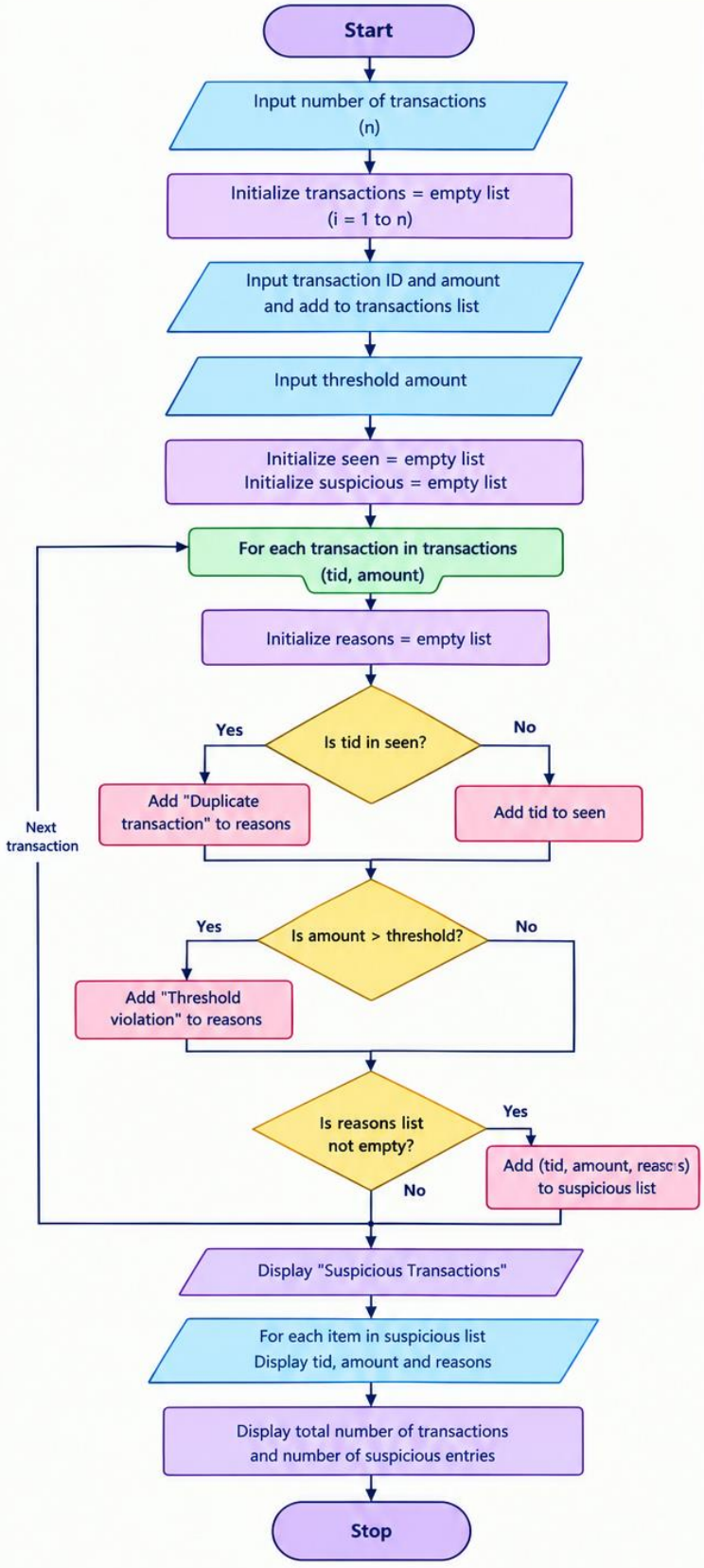
DISPLAY total number of transactions

DISPLAY number of suspicious entries

STOP

Flowchart:

Repeated Transaction Detection (Flow Chart)



Input Details

- Number of transactions (n)
- Transaction ID and amount (n times)
- Threshold amount

Variables Used

- **transactions** – list of (tid, amount)
- **seen** – list of unique transaction IDs
- **suspicious** – list of suspicious entries
- **reasons** – list of reasons for each transaction
- **tid** – transaction identifier
- **amount** – transaction amount
- **threshold** – maximum allowed amount

Conditions

- **Duplicate transaction:** tid already exists in seen.
- **Threshold violation:** amount > threshold.

Suspicious Entry Format

- (tid, amount, reasons)

Output

- List of suspicious transactions (Transaction ID, Amount, Reasons)
- Total number of transactions
- Number of suspicious entries

Python Program:

```
# Repeated Transaction Detection

n = int(input("Enter number of transactions: "))

transactions = []

for i in range(n):
    print("\nTransaction", i + 1)
    tid = input("Enter transaction ID: ")
    amount = float(input("Enter transaction amount: "))
    transactions.append((tid, amount))

threshold = float(input("\nEnter transaction threshold: "))

suspicious = []
seen = []

for transaction in transactions:
    tid, amount = transaction
    reasons = []

    if tid in seen:
        reasons.append("Duplicate transaction")
    else:
        seen.append(tid)
```

```
    if amount > threshold:
        reasons.append("Threshold violation")

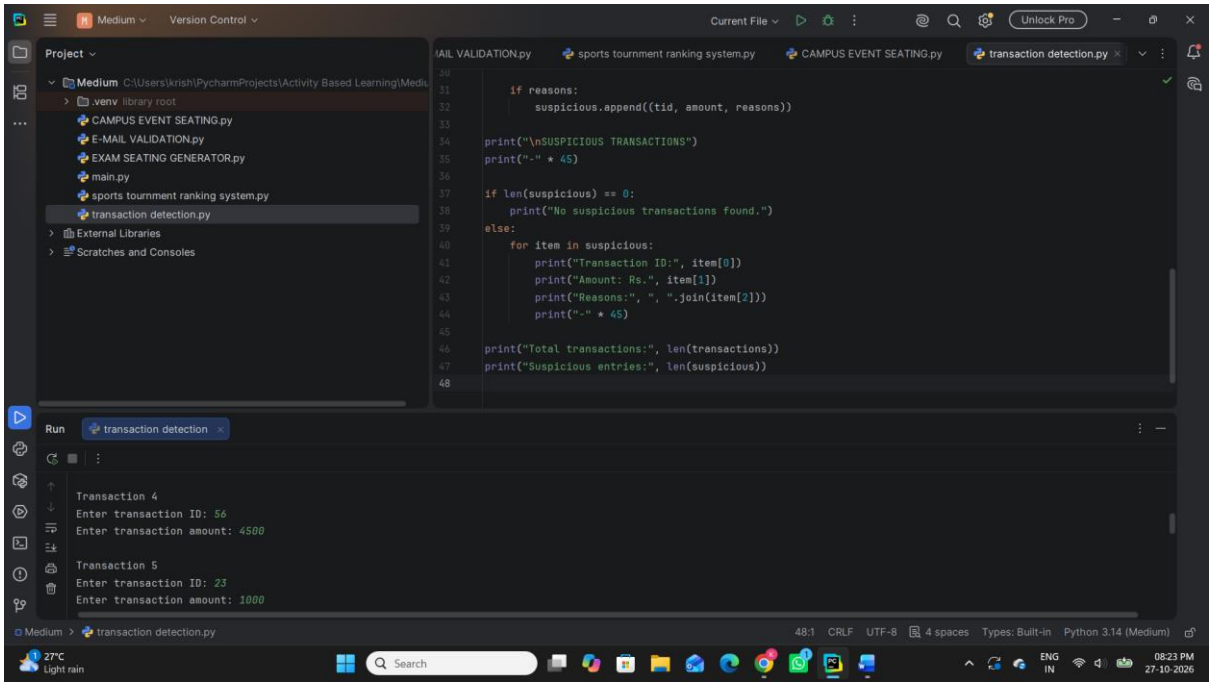
    if reasons:
        suspicious.append((tid, amount, reasons))

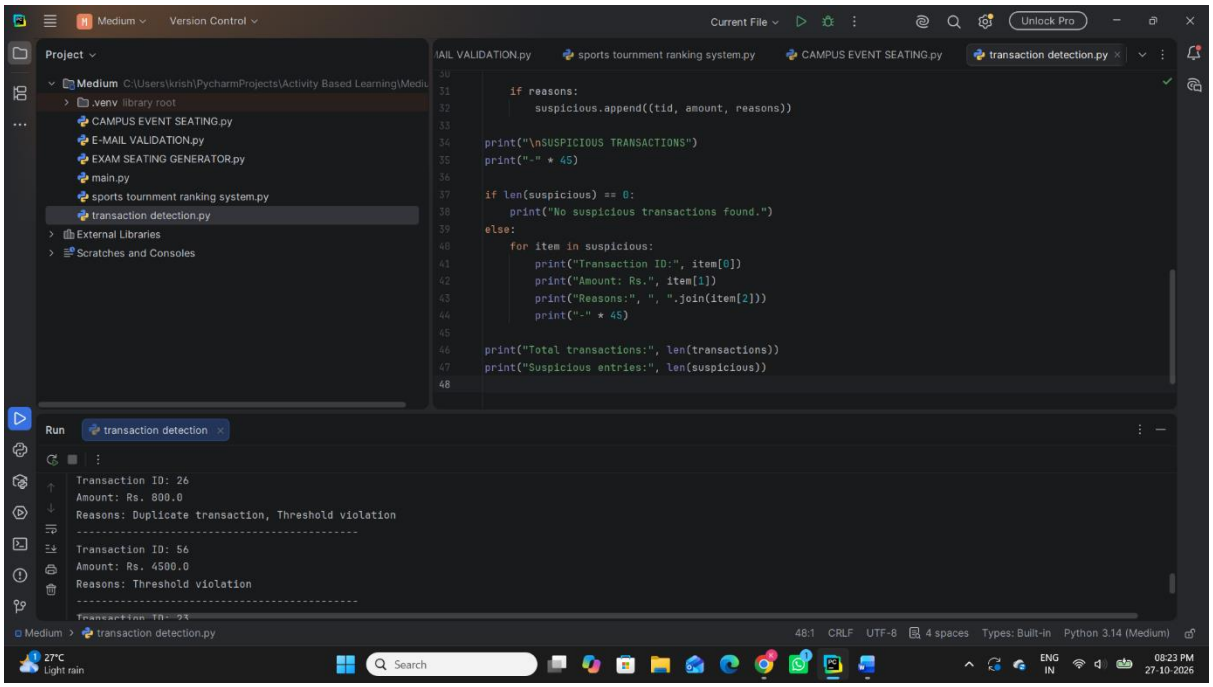
print("\nSUSPICIOUS TRANSACTIONS")
print("-" * 45)

if len(suspicious) == 0:
    print("No suspicious transactions found.")
else:
    for item in suspicious:
        print("Transaction ID:", item[0])
        print("Amount: Rs.", item[1])
        print("Reasons:", ", ".join(item[2]))
        print("-" * 45)

print("Total transactions:", len(transactions))
print("Suspicious entries:", len(suspicious))
```

Output Screen Shots:


Medium
Version Control
Current File
Unlock Pro
Project
Medium C:\Users\krish\PycharmProjects\Activity Based Learning\Mediu
.venv library root
CAMPUS EVENT SEATING.py
E-MAIL VALIDATION.py
EXAM SEATING GENERATOR.py
main.py
sports tournment ranking system.py
transaction detection.py
External Libraries
Scratches and Consoles
AIL VALIDATION.py
sports tournment ranking system.py
CAMPUS EVENT SEATING.py
transaction detection.py
if reasons:
suspicious.append((tid, amount, reasons))
print("\nSUSPICIOUS TRANSACTIONS")
print("-" * 45)
if len(suspicious) == 0:
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for item in suspicious:
print("Transaction ID:", item[0])
print("Amount: Rs.", item[1])
print("Reasons:", ", ".join(item[2]))
print("-" * 45)
print("Total transactions:", len(transactions))
print("Suspicious entries:", len(suspicious))
Run
transaction detection
Transaction 4
Enter transaction ID: 56
Enter transaction amount: 4500
Transaction 5
Enter transaction ID: 23
Enter transaction amount: 1000
Medium
transaction detection.py
48:1
CRLF
UTF-8
4 spaces
Types: Built-in
Python 3.14 (Medium)
27°C
Light rain
Search
ENG
IN
08:23 PM
27-10-2026


Medium
Version Control
Current File
Unlock Pro
Project
Medium C:\Users\krish\PycharmProjects\Activity Based Learning\Mediu
.venv library root
CAMPUS EVENT SEATING.py
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External Libraries
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CAMPUS EVENT SEATING.py
transaction detection.py
if reasons:
suspicious.append((tid, amount, reasons))
print("\nSUSPICIOUS TRANSACTIONS")
print("-" * 45)
if len(suspicious) == 0:
print("No suspicious transactions found.")
else:
for item in suspicious:
print("Transaction ID:", item[0])
print("Amount: Rs.", item[1])
print("Reasons:", ", ".join(item[2]))
print("-" * 45)
print("Total transactions:", len(transactions))
print("Suspicious entries:", len(suspicious))
Run
transaction detection
Transaction ID: 26
Amount: Rs. 800.0
Reasons: Duplicate transaction, Threshold violation
Transaction ID: 56
Amount: Rs. 4500.0
Reasons: Threshold violation
Medium
transaction detection.py
48:1
CRLF
UTF-8
4 spaces
Types: Built-in
Python 3.14 (Medium)
27°C
Light rain
Search
ENG
IN
08:23 PM
27-10-2026